FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – II

JEE (Advanced)-2021

PAPER –1

TEST DATE: 13-06-2021

Time Allotted: 3 Hours **Maximum Marks: 198**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-C.**

  Section-A (01 – 06, 19 – 24, 37– 42) contains **18 multiple** choice questions which have **ONLY ONE CORRECT ANSWER**. Each question carries **+3 marks** for correct answer and **–1 mark** for wrong answer.

  Section-A (07 – 12, 25 – 30, 43 – 48) *this section contains* ***18 multiple*** *choice questions.*
  Each question has FOUR options. ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct answer(s).*
  For each question, choose the option(s) corresponding to (all) the correct answer(s)
  Answer to each question will be evaluated according to the following marking scheme:
  Full Marks **: +4** If only (all) the correct option(s) is (are) chosen:
  Partial Marks **: +3** If all the four options are correct but ONLY three options are chosen;
  Partial Marks **: +2** If three or more options are correct but ONLY two options are chosen and both of which are correct;
  Partial Marks **: +1** If two or more options are correct but ONLY one option is chosen and it is a correct option;
  Zero Marks **: 0** If none of the options is chosen (i. e. the question is unanswered);
  Negative Marks **: –2** In all other cases.

  Section-C (13 – 18, 31– 36, 49 – 54) *contains* ***18 Numerical*** *answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer and* ***0 marks*** *for wrong answer.*

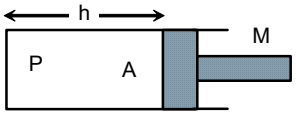
Physics

PART – I

SECTION – A
(One Options Correct Type)

This section contains **06 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A cylindrical piston of mass M slides smoothly inside a long cylinder closed at one end, enclosing a certain mass of a gas. The cylinder is kept with its axis horizontal.



If the piston is slightly compressed isothermally from its equilibrium position, it oscillates simple harmonically, the period of oscillation will be

(A) $T = 2\pi\sqrt{\frac{Mh}{PA}}$

(B) $T = 2\pi\sqrt{\frac{MA}{Ph}}$

(C) $T = 2\pi\sqrt{\frac{M}{PAh}}$

(D) $T = 2\pi\sqrt{MPhA}$

2. A wheel of radius r starts rolling with an angular velocity ω and initial linear velocity V_0 up and inclined smooth plane. The wheel will stop going up in time:

(A) $\frac{V_0}{g\sin\theta}$

(B) $\frac{r\omega}{g\sin\theta}$

(C) $\frac{V_0 - r\omega}{g\sin\theta}$

(D) $\frac{r\omega - V_0}{g\sin\theta}$

3. The binding energy per nucleon of $_5B^{10}$ is 8.0 MeV and that of $_5B^{11}$ is 7.5 MeV. The energy required to remove a neutron from $_5B^{11}$ is (mass of electron and proton are 9.11×10^{-31} kg and 1.67×10^{-27} kg, respectively)

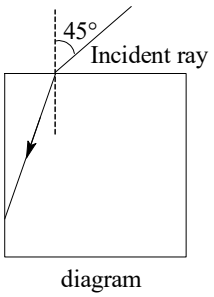
(A) 2.5 MeV
(B) 8.0 MeV
(C) 0.5 MeV
(D) 7.5 MeV

4. A luminous point object is moving along the principal axis of a concave mirror of focal length 12 cm towards it. When its distance from the mirror is 20 cm its velocity is 4 cm/s. The velocity of the image in cm/s at that instant is:

(A) 6, towards the mirror
(B) 6, away from the mirror
(C) 9, away from the mirror
(D) 9, towards the mirror

5. In which of the following cases the potential energy is defined?
   (A) Both conservative and non conservative forces
   (B) Conservative force only
   (C) Non conservative force only
   (D) Neither conservative non conservative force

6. A light ray falls on a square glass slab as shown in the diagram. The index of refraction of the glass, if total internal reflection is to occur at the vertical face, is equal to:



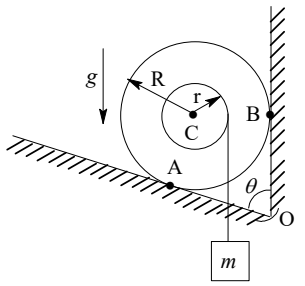
diagram

   (A) $\frac{\left(\sqrt{2}+1\right)}{2}$
   (B) $\sqrt{\frac{3}{2}}$
   (C) $\frac{3}{2}$
   (D) None of these

(One or More than one correct type)

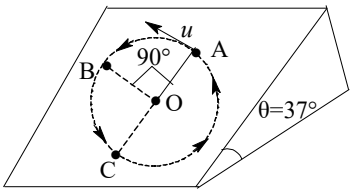
This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

7. A massless spool of inner radius r, outer radius R is placed against a vertical wall and a tilted split floor as shown. A light inextensible thread is tightly wound around the spool through which a mass m is hanging. There exists no friction at point A, while the coefficient of friction between the spool and point B is μ. The angle between the two surfaces is ρ.



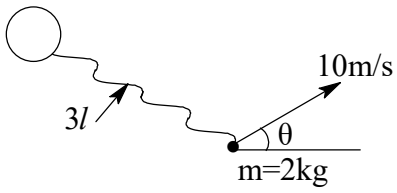
   (A) The magnitude of the force on the spool at B in order to maintain equilibrium is $mg\sqrt{\left(r/R\right)^2+\left(1\text{-}r/R\right)^2\ 1/\tan^2\theta}$
   (B) The magnitude of the force on the spool at B in order to maintain equilibrium is $mg\left(1\text{-}r/R\right)1/\tan\theta$
   (C) The minimum value of μ for the system to remain in equilibrium is $\cot\theta/\left(R/r\right)\text{-}1$
   (D) The minimum value of μ for the system to remain in equilibrium is $\tan\theta/\left(R/r\right)\text{-}1$

8. A small sphere of mass m is connected by a string to a nail at O and moves in a circle of radius r on the smooth plane inclined at an angle θ with the horizontal. If the sphere has a velocity u at the top position A. Mark the correct options.



   (A) Minimum velocity at A so that the string does not get slack instantaneously is $\sqrt{3/5gr}$
   (B) Tension at B is 9mg/5 if the sphere has the required velocity as in option (A).
   (C) Tension at C is 18mg/5 in situation of option (B)
   (D) None of these

9. A string of length $3l$ is connected to a fixed cylinder whose top view is shown in figure. The string is initially slack. The other end of the string (connected to a marble) is moving at a constant velocity of 10 m/s as shown. The string will get stretched at some instant and impulsive tension occurs in the string. If hinge is exerting a force of 40000 N for 0.25 ms on the cylinder to bear up the impact of impulsive tension, then mark the correct statements. (Take string to be light, breaking tension of the string is $2\times10^5\text{N}$)



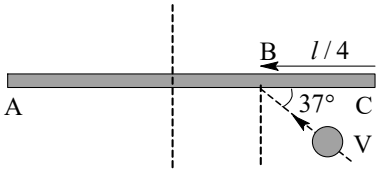
(A) The angle made by the velocity of marble with the length of string when it is just stretched is $60°$

(B) The marble will move in a circular path of varying radius with constant speed of $5\sqrt{3}$ m/s, after the string is taut

(C) To answer above two options, the value of θ must be given

(D) The string will break is impulse duration is less than 0.05 min

10. A rod AC of length l and mass m is kept on a horizontal smooth plane. It is free to rotate and move. A particle of same mass m moving on the plane with velocity v strikes the rod at point B making angle 37° with the rod. The collision is elastic. After collision.



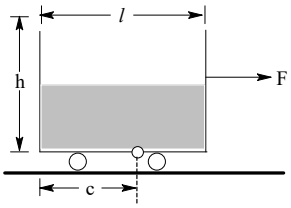
(A) The angular velocity of the rod will be 72/55 v/l

(B) The centre of the rod will travel a distance $\pi l/3$ in the time in which it makes half rotation

(C) Impulse of the impact force is $24\text{m v}/55$

(D) None of these

11. A rectangular vessel of dimension $(l\times b\times h)$ and mass M contains a liquid of density ρ. The vessel has an orifice at its bottom at a distance c from the rear wall as shown in the figure.



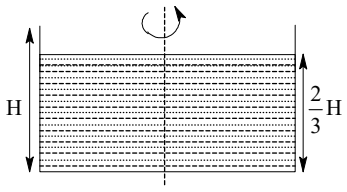
(A) The maximum volume of the water that can be stored when the vessel is accelerated is $hcb/2$

(B) The maximum volume of the water that can be stored when the vessel is accelerated is $hlb/2$

(C) Force F that must be applied when maximum water stored is $\left[M+hcb\rho/2\right]hg/c$

(D) Force F that must be applied when maximum water stored is $\left[M+hcb\rho/2\right]lg/c$

12. A circular cylinder of radius R and height H is filled with water to a height $\left(\frac{2}{3}\right)\text{H}$. It starts rotating about its axis with constantly increasing angular speed. Choose the correct alternatives.



(A) At all speeds, shapes of the free surface is parabolloid

(B) The free surface touches first the brim of the cylinder and then the base of the cylinder.

(C) The free surface cannot touch the base without spilling water.

(D) The free surface touches the brim as well as the base at the same instant.

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

13. A neutral atom of atomic mass number 100 which is stationary at the origin in gravity free emits an α-particle (A) in z-direction. The product ion is P. A uniform magnetic field exists in the x-direction. Disregard the electromagnetic interaction between A and P. If the angle of rotation of A after which A and P will meet for the first time is $6n\pi/25$ radius, what is the value of n?

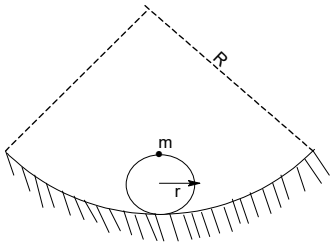
14. A neutral particle of mass 28g is at rest in a uniform magnetic field $\text{B}=\frac{200}{3}\text{T}$. At $\text{t}=0$, particle decays into two particle of equal mass and one of them having charge $3\mu\text{C}$. Both of them move in separate paths lying in the plane perpendicular to magnetic field B. At later time t, the particle collides. (Neglecting the interaction force). Calculate $\frac{\text{t}}{100}$.

15. A vertical capillary tube with inside radius 0.25 mm is submerged into water so that the length of its part protruding over the water surface is equal to 25 mm. Surface tension of water is $73\times10^{-3}\text{N/m}$ and angle of contact is zero degree for glass and water, acceleration due to gravity is $9.8\,\text{m/s}^2$. Then value of 10R approximately (in mm) is (where R is radius of meniscus and h is height of water in capillary tube)

16. A wire of length L is shaped into a circle and then bent in such a way that the two semi-circles are perpendicular. The magnetic moment of the system when current flows through the system is $\left(\frac{\sqrt{2}iL^2}{8\pi}\right)n$, then value of n is

17. If the sound heard by observer, whose equation is given as $y=8\sin10\pi t\cos200\pi t$ at $x=0$, then number of beat frequency heard by observer is 2k. Then the value of k is___

18. A ring of radius r and mass 2m, rests on a rough circular track of radius R.A small point mass of mass m is mounted on its periphery. If friction is sufficient to avoid any slipping then the minimum ratio of $\frac{r}{R}$ is



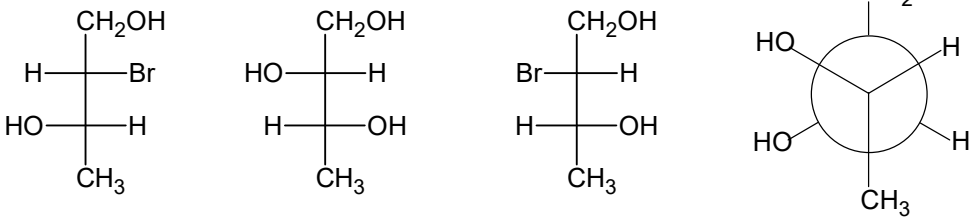
Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **06 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

19. Consider the following statements regarding the given projections.

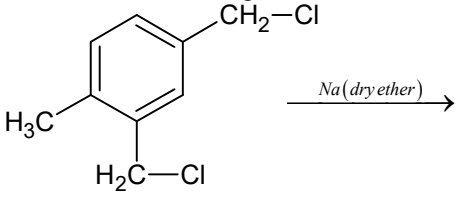


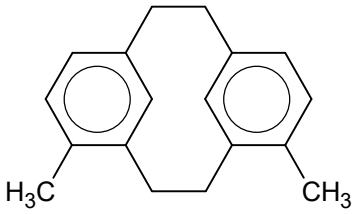
(A) (B) (C) (D)

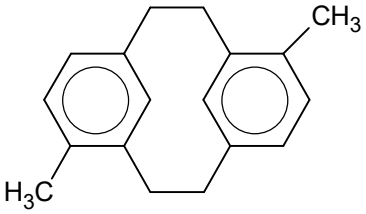
(1) (A) and (C) are diastereoisomers.
(2) (D) is the Newmann projection of (B).
(3) (B) may be named as thereo-1, 2, 3-butanetriol.
Which of the above statements are correct?

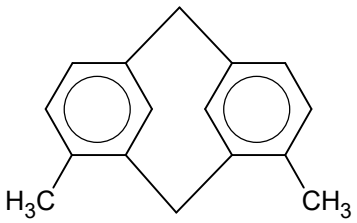
(A) 1, 2 & 3
(B) 1& 2
(C) 2& 3
(D) 3

20. Product obtained in given Wurtz reaction is:



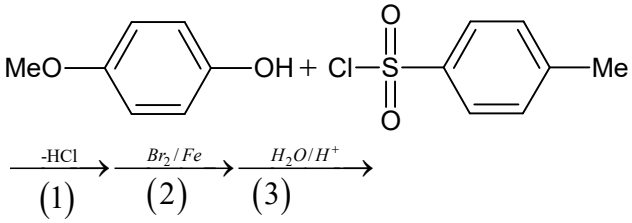
(A)


(B)


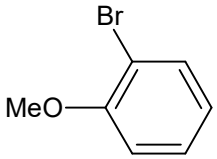
(C)


(D) Both (A) and (B)

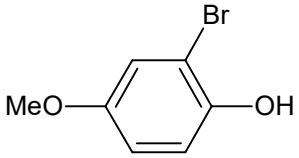
21. The major product is:



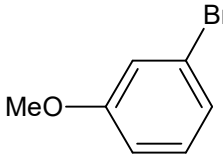
(A)



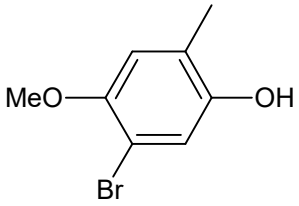
(B)



(C)

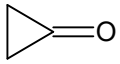


(D)

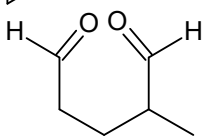


22. Compound $P\left(C_6H_{10}\right)$ does not have any geometrical isomer, ON ozonolysis, two products $R\left(C_3H_4O\right)$ and $Q\left(C_3H_6O\right)$ are formed. R gives negative iodoform test while Q responds positively towards $I_2/NaOH$ solution. S, another isomer of P is an unsymmetrical alkene and on ozonolysis produces $T\left(C_6H_{10}O_2\right)$ which also gives a yellow precipitate with $I_2/NaOH$ solution and also gives positive test with Tollen's reagent. Which of the following does not represent any of the molecules amongst P, Q, R, S & T?

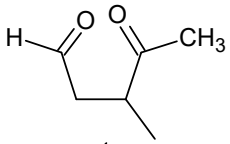
(A)



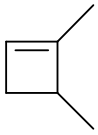
(B)



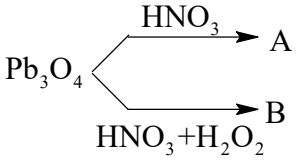
(C)



(D)



23. In the following reactions, the Pb compounds A and B are respectively



(A) $Pb(NO_3)_2 + PbO_2$ and $Pb(NO_3)_2$

(B) $Pb(NO_3)_2$ and $Pb(NO_3)_2$

(C) PbO_2 and $Pb(NO_3)_2$

(D) $Pb(NO_3)_2$ and $PbO_2 + Pb(NO_3)_2$

24. Which is not correct regarding the Holme's signal?

(A) it is PH_3 which catches fire on contact with air, spontaneously.

(B) it is P_2H_4 which catches fire on contact with air, spontaneously.

(C) all these PH_3, C_2H_2 and P_2H_4 are formed.

(D) The gases formed all can burn to form signal.

(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

25. Which of the following statements is/are true regarding N_2O

(A) It is used as anaesthetic by dentists

(B) Bond orders are fractional for N – N and N – O bonds.

(C) It reacts with water to give HNO_3 and HNO_2.

(D) It is very good supporter of combustion.

26. The colour of a compound may be due to:

(A) Polarization.

(B) d-d transition.

(C) f-f transition.

(D) Charge transfer spectra.

27. Which of the following statements are correct?

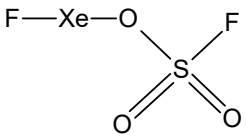
(A) The electron affinity of Si is greater than that of C.

(B) BeO is amphoteric while B_2O_3 is acidic.

(C) The ionization energy of Tl is less than that of Al.

(D) The ionization energy of elements of Cu-group is less than that of the respective elements of Zn – group.

28. Which of the following statements is/are correct for $XeFOSO_2F$?



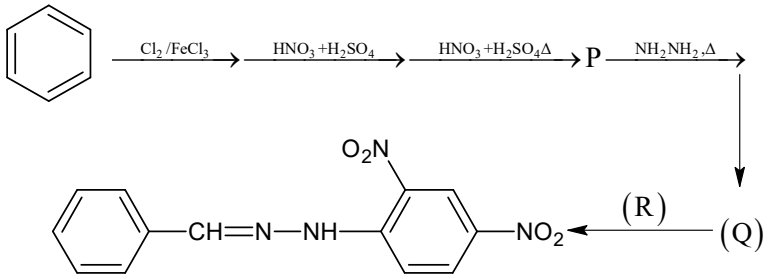
(A) The number of sp^3 hybridized central atoms is 3.

(B) The number of identical S – O bond is 2.

(C) Maximum number of atoms that may lie in one plane is 5.

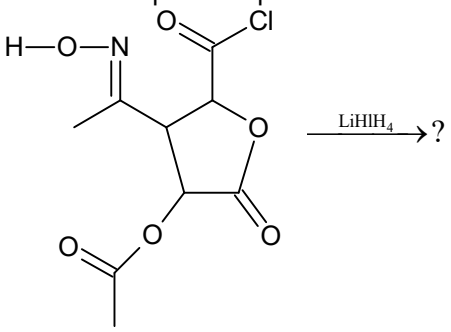
(D) All atoms are laying in the same plane.

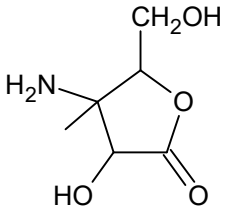
29. The correct statement/s about the following reaction sequence is/are:

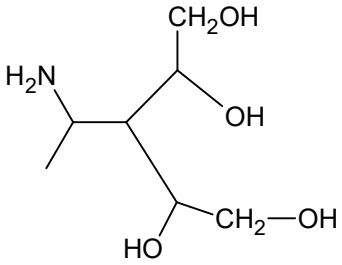


(A) 'R' gives an aldol condensation reaction on heating with NaOH solution.
(B) Compound 'R' with HCHO and conc. NaOH forms sodium formate as one of the product.
(C) Base catalysed haloform reaction involves formation of carbanion.
(D) Acid catalysed halogenation of CH_3COCH_3 involves enol formation.

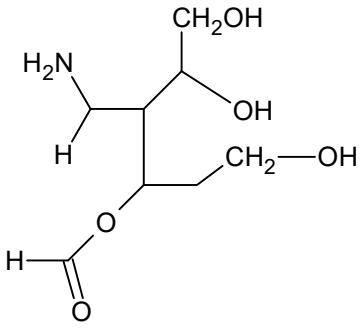
30. Find out the possible products in the following reaction:



(A) 

(B) 

(C) CH_3CH_2OH

(D) 

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

31. What mass of ^{14}C with $t_{1/2} = 5730$ year has activity equal to one curie?

32. A container of volume V L contains an equilibrium mixture that consists of 2 mol each of gaseous PCl_5, PCl_3 and Cl_2 at 3 atm and $T\ K$. Some Cl_2 is added until the volume is double keeping P and T constant. Calculate moles of Cl_2 added $PCl_5 \rightarrow PCl_3 + Cl_2$.

33. An aqueous solution of an acid is so weak that it can be assumed to be practically unionised, boiled at $100.4°C$. $25\,mL$ of this solution was neutralised by $38.5\,mL$ of $1N$ solution of $NaOH$. Calculate basicity of the acid if K_b for water is $0.52\,K\,mol^{-1}kg$. Assume molality equal to molarity.

34. E.M.F. diagram for some ions is given as:

$$FeO_4^{2-} \xrightarrow{E°=+2.20V} Fe^{3+} \xrightarrow{E°=+0.77V} Fe^{2+}$$

$$Fe^{2+} \xrightarrow{E°=-0.445V} Fe°$$

Determine the value of $E°_{FeO_4^{2-}/Fe^{2+}}$

35. An electric current is passed through two solutions of $(i)\ AgNO_3$ and (ii) a solution of 10g $CuSO_4.5H_2O$ crystals in 500 mL H_2O, platinum electrodes being used in each case. After 30 minute it is found that 1.307g Ag has been deposited. What was the conc. of Cu expressed in g of Cu per litre in solution after electrolysis?

36. A 1.100g sample of copper ore is dissolved and the Cu^{2+} of it is treated with excess KI. The liberated I_2 requires $12.12\,mL$ of $0.10M$ $Na_2S_2O_3$ solution for titration. What is % copper by mass in the ore?

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **06 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

37. Tangents at P, Q, R on a circle of radius r form a triangle whose sides are 3r, 4r, 5r then $PR^2 + RQ^2 + QP^2$ =

(A) $\frac{84}{5}r^2$

(B) $\frac{42}{5}r^2$

(C) $\frac{44r^2}{5}$

(D) None of these

38. A parabola $y = ax^2 + bx + c$ crosses the x-axis at $(\alpha,0)$ and $(\beta,0)$ both to the right of the origin. A circle also passes through these two points. The length of a tangent from the origin to the circle is:

(A) $\sqrt{\frac{bc}{a}}$

(B) ac^2

(C) $\frac{b}{a}$

(D) $\sqrt{\frac{c}{a}}$

39. The expression $(x+2y)^{20}$ contain two terms with the same coefficient, $k\,x^a y^b$ and $k\,x^{a+1}y^{b-1}$, then a is

(A) 8

(B) 5

(C) 7

(D) 6

40. The area bounded by y = f(x), $y = f^{-1}(x)$ and line x + y = 7/6 where $f(x) = \frac{x}{3} - \frac{x^2}{6}\ \forall\ x \in [0, 1]$ is

(A) $\frac{7}{16}$

(B) $\frac{31}{72}$

(C) $\frac{21}{16}$

(D) $\frac{31}{36}$

41. The minimum distance between the curves in Argand plane $\left|\frac{z-1}{z-2}\right| = 2$ and $\left|\frac{z-2}{z-1}\right| = 2$

(A) 1
(B) 0
(C) 2/3
(D) 1/3

42. Least value of expression $\frac{1}{2bx - \left(x^2 + b^2 + \sin^2 x\right)}, x \in [-1,0], b \in [2,3]$ is

(A) 1/4
(B) –1/4
(C) 1/2
(D) None of these

(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

43. Equation of a line that touches the curve $y = x|x|$ and $x^2 + (y - 2)^2 = 4$, where $x \neq 0$, is

(A) $y = 4\sqrt{3}x - 12$
(B) $y = -4\sqrt{5}x + 20$
(C) $y = -4\sqrt{5}x - 20$
(D) $y = 4\sqrt{5}x + 20$

44. K subsets of a set $A \equiv \{1, 2, 3, \ldots, n\}$ are chosen at random with replacement. The probability that their intersection is

(A) empty is $\left(\frac{1}{2}\right)^n$ for K = 2

(B) empty is $\left(\frac{3}{4}\right)^n$ for K = 2

(C) empty is $\left(\frac{7}{8}\right)^n$ for K = 3

(D) singleton is $n\left(\frac{7}{8}\right)^{n-1}$ for K = 3

45. An extreme value of $y = 4\sin^2 x + 3\cos^2 x - 24\sin\frac{x}{2} - 24\cos\frac{x}{2}$, where $0 \le x \le \frac{\pi}{2}$, is

(A) –21
(B) $4 + \sqrt{2}$
(C) 4
(D) $4(1 - 6\sqrt{2})$

46. Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and let $\vec{r}$ be a variable vector such that $\vec{r} \cdot \hat{i}$, $\vec{r} \cdot \hat{j}$ and $\vec{r} \cdot \hat{k}$ are positive integers. If $\vec{r} \cdot \vec{a} \le 12$, then the number of values of $\vec{r}$ will be

(A) ${}^{12}C_3$
(B) ${}^{13}C_4$
(C) ${}^{13}C_9$
(D) ${}^{11}C_8 + {}^{11}C_9$

47. Let $f(x) = \lim_{n\to\infty} \frac{\tan x^2 + (1+x)^n \sin x}{x^2 + (1+x)^n}, n \in N$ then

(A) f(x) is continuous at x = 0

(B) $\lim_{x\to 0} f(x) = 0$

(C) $\lim_{x\to 0^+} f(x) = 0$

(D) $\lim_{x\to 0^-} f(x) = 1$

48. Let functions $f, g : R \to R$ defined by $f(x) = 3x - 1 + |2x+1|$, $g(x) = \frac{1}{5}(3x + 5 - |2x+5|)$. Then which relation between f and g is/are true

(A) $gof = fog$

(B) $(fog)^{-1} = gof$

(C) $(fof) = gog$

(D) $(fof)^{-1} = gog$

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

49. $\int_0^3 (x^2+1)\, d([x])$ where [x] is the greatest integer less than or equal to x_____

50. The number of real roots of the equation $54x^4 - 36x^3 + 18x^2 - 6x + 1 = 0$ is_____

51. If the equations $x^2 - 3x - a_i = 0$ has integral roots $\forall\ a_i \in N$ and $a_i \le 300$ then $\sum a_i$ = _____

52. If a 9-digit numbers is formed using the digits 1 to 9 without repetition then the probability that they will be divisible by 11 is p/q then p + q equals _____ (p and q are prime to each other)

53. If $\cos\frac{\pi}{n+1}$ is a root of the equation $8x^3 + 4x^2 - 4x - 1 = 0$. Then n = _____ ($n \in N$).

54. The area bounded by the curve $f(x) = x + \sin x$ and its inverse between the ordinates x = 0 to $x = 2\pi$ is 2^k, $k \in N$, then k = _____